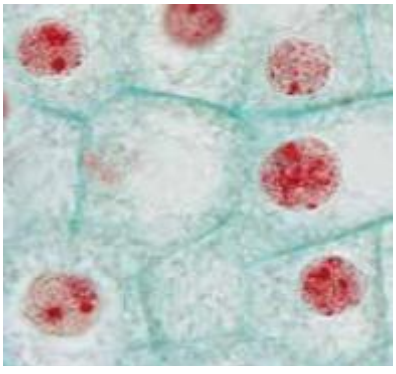
Multiple-choice questions

|!|BCEMC0120000001|!|

The photomicrograph below shows some cells at a certain stage in the cell cycle.



(x1000)

Which of the following combinations correctly matches the type of cell and its stage in the cell cycle?

	Type of cell	*Stage in the cell cycle*
A.	plant cell	prophase
B.	animal cell	prophase
C.	plant cell	interphase
D.	animal cell	telophase

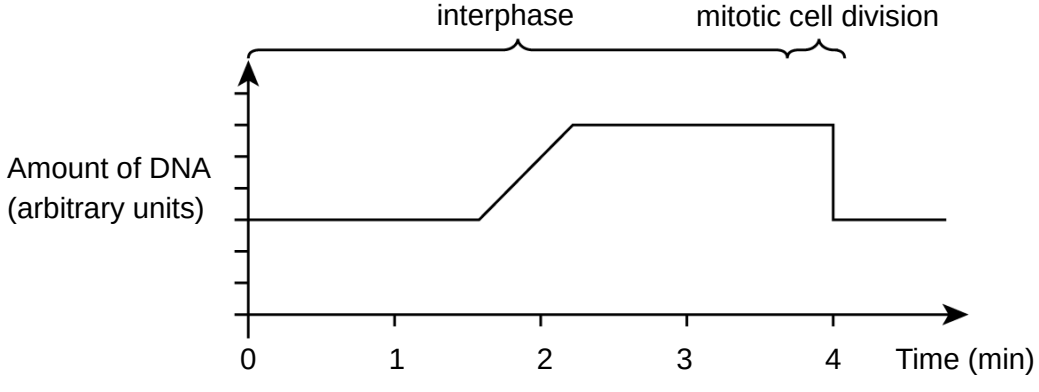
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0120000002|!| ★★

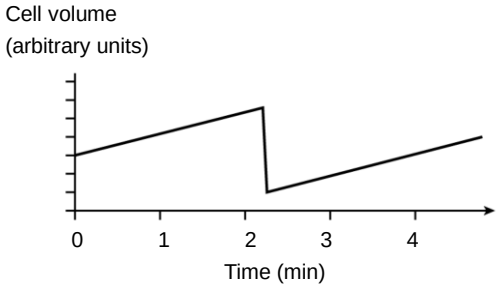
The graph below shows the change in the amount of DNA during a cell cycle.



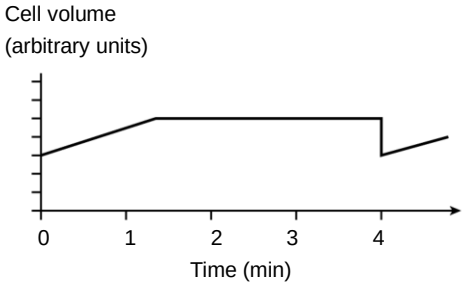
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Which of the following correctly shows the change in the cell volume during the cell cycle?

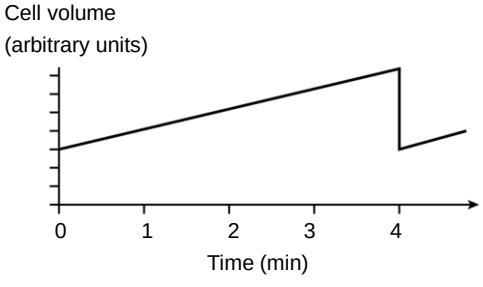
A.



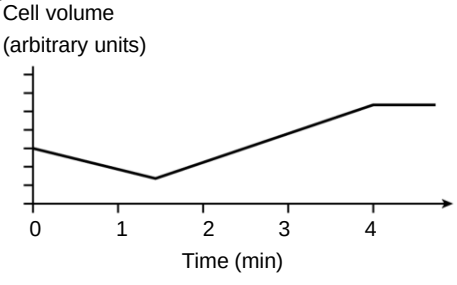
B.



C.



D.



##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0120000004|!| ★★

During which stage in the cell cycle does the surface area to volume ratio of the cell decrease?

A. interphase
B. prophase
C. metaphase
D. cytokinesis

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0120000005|!| ★

At which stage of meiosis do the sister chromatids of each chromosome separate?

A. metaphase I
B. anaphase I
C. metaphase II
D. anaphase II

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0120000006|!| ★★

A fruit fly has 8 chromosomes in a diploid cell. Which of the following combinations correctly shows the chromosome numbers of each daughter cell after the first and second meiotic divisions?

	After the first meiotic division	***After the second meiotic division***
A.	8	8
B.	4	4
C.	8	4
D.	4	2

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0120000007|!|

Generally, cells spend most time in

A. prophase.
B. telophase.
C. interphase.
D. cytokinesis.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000008|!|

During which stage of meiosis do the homologous pairs of chromosomes separate?

A. prophase I
B. anaphase I
C. anaphase II
D. telophase II

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000009|!|

The following events occur in mitosis.

(1) The spindle fibres attach to the centromeres.
(2) DNA coils up tightly into chromosomes.
(3) Sister chromatids move to the opposite poles of the cell.

A. (1) → (2) → (3)
B. (2) → (3) → (1)
C. (2) → (1) → (3)
D. (1) → (3) → (2)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000010|!|

Which of the following statements about the cell cycle are correct?

(1) The cell cycle consists of both interphase and mitosis.
(2) Interphase typically lasts for about 25% of the cell cycle.
(3) DNA replication takes place in interphase.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000011|!|

The following photomicrograph shows various stages of mitotic cell division.



How many cells are in metaphase?

A. 1

B. 2

C. 4

D. 5

##C##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0120000012|!| ★

During which stage of the cell cycle are the numbers of mitochondria increased in a cell?

A. prophase
B. metaphase
C. interphase
D. cytokinesis

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0120000013|!| ★ **STSE**

Chemical X can be used as an anti-cancer drug. It works by stopping the chromatids from separating during mitosis. Which phase would the cancer cells reach and then stop dividing?

A. anaphase
B. metaphase
C. prophase
D. telophase

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0120000014|!| ★

Independent assortment of homologous chromosomes could be beneficial to the survival of a species because

A. it reduces the number of chromosomes in a diploid parent cell by half.
B. it produces different genetic combinations in gametes and increases genetic variation.
C. it keeps the chromosome number of the species constant generation after generation.
D. it ensures that the daughter cells are genetically identical to the parent cell.

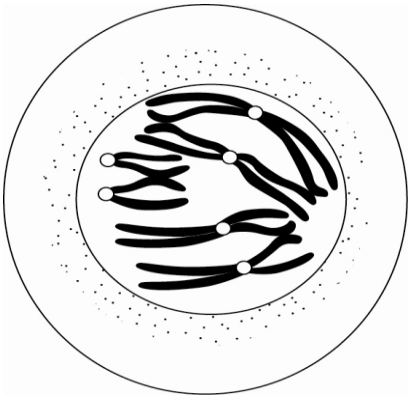
##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000016|!| ★

The following diagram shows a cell undergoing meiosis.



In which stage of meiosis is the cell?

A. prophase I
B. metaphase I
C. anaphase I
D. telophase I

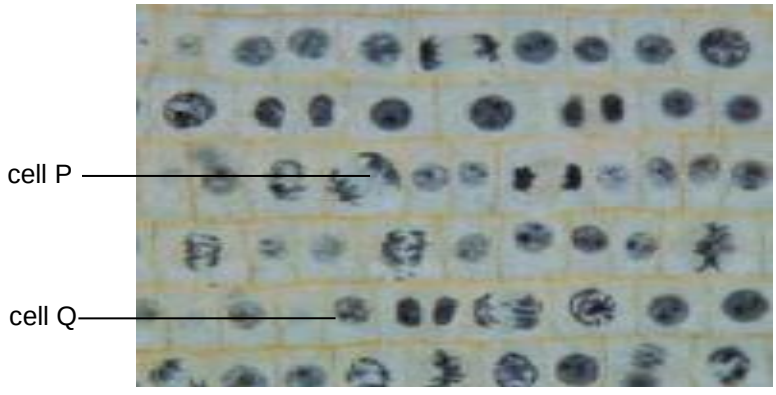
□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000017|!| ★

The photomicrograph below shows the longitudinal section of an onion root tip.



Which of the following correctly matches cells P and Q with their stages of cell cycle?

	Cell P	***Cell Q***
A.	metaphase	prophase
B.	anaphase	metaphase
C.	interphase	telophase
D.	anaphase	interphase

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000018|!|

Which of the following comparisons between mitotic cell division and meiotic cell division is ***incorrect***?

	Mitotic cell division	***Meiotic cell division***
A.	occurs in growing regions	occurs in reproductive organs
B.	daughter cells are diploid	daughter cells are haploid
C.	produces two body cells per division	produces two gametes per division
D.	involves one nuclear division	involves two nuclear divisions

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000019|!| ★

When cytoplasmic division begins in a plant cell, what are formed between the two daughter nuclei?

(1) spindle fibres
(2) cell wall
(3) cell membrane

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000020|!|

What is the outcome when a cell undergoes meiotic cell division?

A. two body cells
B. four body cells
C. two gametes
D. four gametes

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000021|!|

Which of the following processes is ***not*** involved in mitosis?

A. coiling up of chromatin into chromosomes
B. pairing of homologous chromosomes
C. separation of sister chromatids
D. disintegration of the nuclear membrane

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000022|!| ★ **Cross topic**

Which of the following organisms can reproduce asexually by mitotic cell division?

(1) *Amoeba*
(2) *Paramecium*
(3) potato

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000023|!|

What is the chromosome number of a human sperm?

A. 23
B. 26
C. 43
D. 46

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000024|!| ★

How many pairs of chromosomes are found in a daughter cell formed by a stem cell in the human skin?

A. 13
B. 23
C. 26
D. 46

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000025|!|

Which of the following events happens when cytoplasmic division begins in an animal cell?

A. The cell membrane constricts around the middle of the cell.
B. A cell plate is formed around the middle of the cell.
C. Spindle fibres appear.
D. The nuclear membrane disintegrates.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000026|!|

During which stage does replication of DNA occur in a cell?

A. interphase
B. prophase
C. anaphase
D. telophase

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000027|!|

Which of the following does ***not*** involve mitotic cell division?

A. asexual reproduction
B. gamete formation
C. formation of a tumour
D. healing of wound

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000028|!| ★

Which of the following comparisons between chromatin and chromatids are correct?

	Chromatin	***Chromatids***
(1)	appears when a cell is not dividing	appear when a cell is about to divide
(2)	do not contain proteins	contains proteins
(3)	appears as long and fine strands	appear as highly coiled strands

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

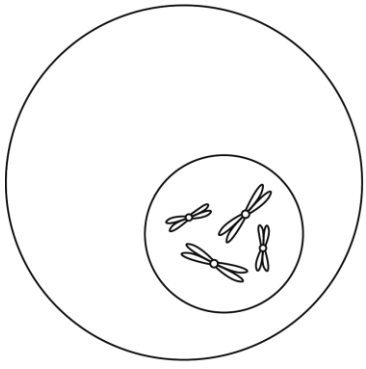
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000029|!| ★

The following diagram shows a body cell.



Which of the following statements about the cell is correct?

A. The cell is in interphase.
B. The cell is synthesizing new organelles.
C. The chromosomes are going to uncoil and become chromatin.
D. The homologous chromosomes are about to pair up.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000030|!|

Which is the correct sequence of the stages of mitosis?

(1) Chromosomes line up in the middle of the cell.
(2) DNA molecules coil up into chromosomes.
(3) The nuclear membrane breaks down.
(4) Sister chromatids of each chromosome are pulled apart towards opposite poles of the cell.
(5) The chromosomes uncoil to become chromatin.

A. (5), (2), (3), (1), (4)
B. (2), (3), (1), (4), (5)
C. (3), (2), (1), (4), (5)
D. (3), (5), (4), (2), (1)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000031|!| ★

Which of the following types of cells can undergo mitotic cell division?

(1) red blood cells
(2) B lymphocytes
(3) skin cells

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0120000032|!| ★

Which of the following statements about different stages of meiosis is/are correct?

(1) Crossing over takes place during prophase I.
(2) Sister chromatids separate during anaphase II.
(3) Homologous chromosomes pair up during metaphase II.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000033|!| ★

Which of the following types of cells is diploid?

A. xylem vessel
B. ovum
C. mature red blood cell
D. white blood cell

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0120000034|!|

How many offspring are produced by two *Paramecia* that have carried out mitotic cell division once?

A. 2
B. 4
C. 8
D. 16

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMA0120000035|!| ★★

How many pairs of homologous chromosomes can be found in a human sperm?

A. 0
B. 1
C. 23
D. 46

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000036|!|

Which of the following statements about chromosomes is/are correct?

A. Each chromosome appears as two sister chromatids during telophase.
B. At the end of mitosis, chromatin coils up to become chromosome.
C. Chromosome appears as chromatin during interphase.
D. The two members of a pair of homologous chromosomes are genetically identical.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000037|!| ★

Which of the following statements about mitotic cell division is/are ***incorrect***?

(1) DNA only replicates once during prophase.
(2) After mitosis, the number of chromosomes in the daughter cells is halved.
(3) The cells in the apical meristem of a plant divide by mitotic cell division.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000038|!|

After mitotic cell division, a parent cell with 20 chromosomes produces

A. 2 daughter cells each containing 10 chromosomes.
B. 2 daughter cells each containing 20 chromosomes.
C. 4 daughter cells each containing 10 chromosomes.
D. 4 daughter cells each containing 20 chromosomes.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000039|!|

Which of the following indicate(s) the significance of mitotic cell division?

(1) for growth and repair
(2) for creating genetic variation in offspring
(3) for asexual reproduction

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0120000040|!|

In humans, which of the following processes involve(s) mitotic cell division?

(1) production of white blood cells
(2) synthesis of salivary amylase
(3) production of pancreatic juice

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0120000041|!|

Which of the following statements about the daughter cells formed after meiotic cell division in humans are correct?

(1) They have half the number of chromosomes of their parent cells.
(2) They can be found in the ovaries and testes.
(3) They form gametes.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000042|!|

Which of the following show(s) the significance of meiotic cell division?

(1) The diploid number of chromosomes can be restored after fertilization.
(2) The chromosome number of a species can be kept constant over many generations.
(3) It produces gametes with new genetic combinations.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

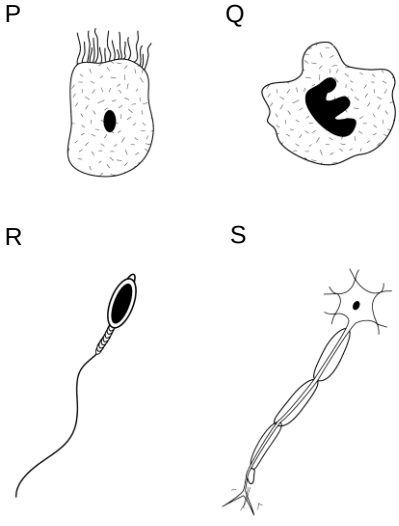
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000043|!|

The diagrams below show a variety of human cells.



Which cell has only 23 chromosomes?

A. P
B. Q
C. R
D. S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000044|!| ★

Which of the following types of human cells contains the largest amount of DNA?

A. red blood cell
B. ovum
C. sperm
D. muscle cell

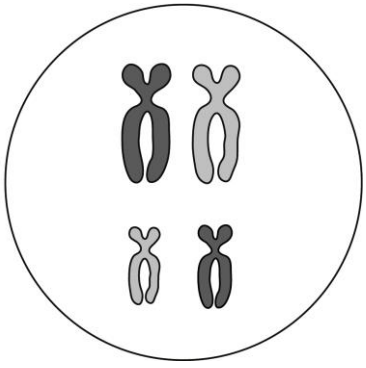
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0120000045|!| ★

The following diagram shows a stage of cell division in a human cell. (Only two pairs of chromosomes are shown.)



Which of the following information about the cell can be deduced from the diagram?

(1) It can be found in the sex organ.
(2) It is carrying out mitosis.
(3) It can produce daughter cells with genetic variation.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000046|!|

Which of the following is a haploid cell in flowering plants?

A. xylem vessel
B. a mesophyll cell
C. root hair cell
D. an ovum

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0120000047|!| ★★

Which of the following statements about crossing over is ***incorrect***?

A. It involves the exchange of genetic material between sister chromatids of a chromosome.

B. It happens during prophase I of meiosis.

C. It does not happen in mitotic cell division.

D. It creates new combinations of genetic make-up in gametes.

□

##A##

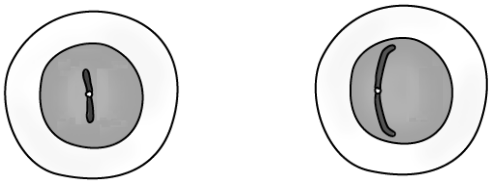
Skill assessed	Mark
Understanding basic concepts	1

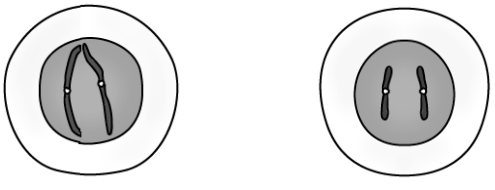
|!|BCEMB0120000048|!| ★

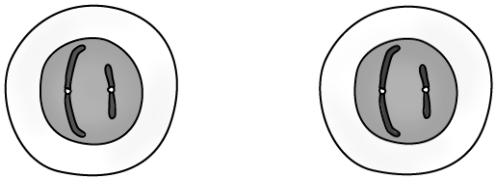
Assume that there are two pairs of chromosomes in a parent cell, as shown below.



Which of the following daughter cells are produced after the cell has carried out mitotic cell division?

A. 

B. 

C. 

D.  

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000049|!| **Cross topic**

Which of the following statements about mitotic cell division in plant cells are correct?

(1) New combinations of genetic make-up are created in the daughter cells.
(2) A cell plate is formed between the two newly formed nuclei.
(3) This division occurs in the apical meristem of the plant.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000050|!| ★

After the first meiotic division, which of the following are formed from one human cell?

A. two haploid cells
B. four haploid cells
C. two diploid cells
D. four diploid cells

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0120000051|!|

Which of the following statements about interphase is ***incorrect***?

A. During interphase, the genetic material appears as chromatin.

B. Interphase is the first stage of mitosis.

C. Synthesis of new organelles occurs during interphase.

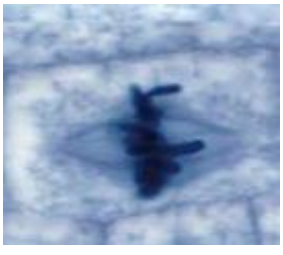
D. DNA replication occurs during interphase.

##B##

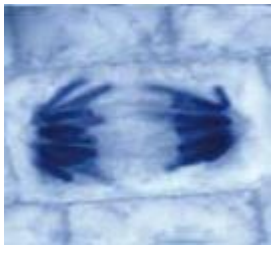
Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0120000052|!| ★

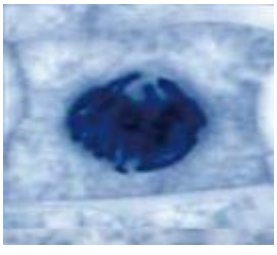
The following photomicrographs show the different stages of a cell division in root cells.



X



Y



Z

Which of the following statements about these stages is ***incorrect***?

A. The chromosomes are attached to the spindle fibres at stage X.

B. Two members of each homologous pair separate at stageY.

C. Stage Y happens after stage X.

D. The nuclear membrane is broken down at stage Z.

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0120000055|!| ★ **Cross topic**

Which of the following does ***not*** involve mitotic cell division?

A. healing of wound
B. growth of bone
C. reproduction of *Amoeba*
D. increase in the size of a root cell

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0120000056|!|

Which of the following statements about meiotic cell division is ***incorrect***?

A. A cell carrying out meiotic cell division divides into four daughter cells.
B. In humans, sperm production involves meiotic cell division.
C. Meiotic cell division is important in the reproduction of bacteria.
D. Meiotic cell division produces gametes which are genetically different.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0120000057|!|

Which of the following statements about the process of meiosis is ***incorrect***?

A. Homologous pairs line up along the equatorial plane at metaphase I.
B. Crossing over may occur at prophase I.
C. The nuclear membrane disintegrates twice during meiosis.
D. The sister chromatids separate at anaphase I.

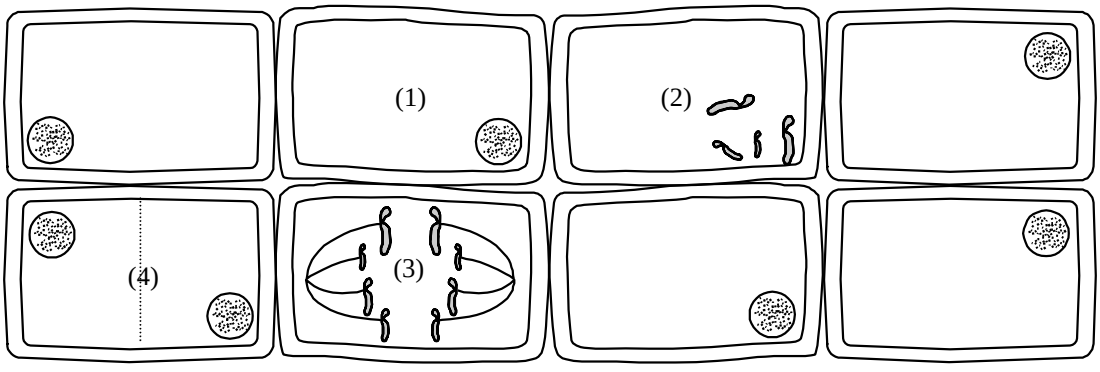
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0120000062|!| ★

The following diagram shows the cells in the apical meristem of a plant. These cells are in different stages of cell division.



Which of the following shows the correct order of the different stages of cell division?

A. (1) → (2) → (3) → (4)
B. (1) → (4) → (3) → (2)
C. (2) → (3) → (1) → (4)
D. (4) → (1) → (3) → (2)

##A##

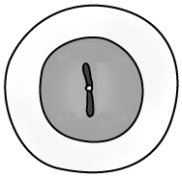
Skill assessed	Mark
Analysing data	1

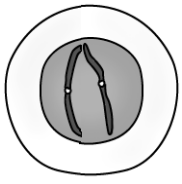
|!|BCEMB0120000063|!| ★

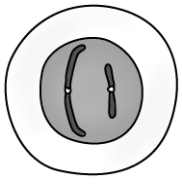
Assume that there are two pairs of chromosomes in a parent cell, as shown below.



Which of the following daughter cells is produced after the cell has carried out meiotic cell division?

A. 

B. 

C. 

D. 

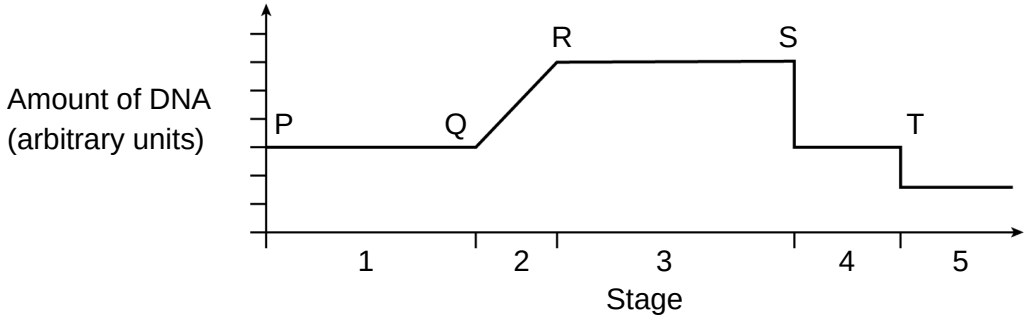
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000064|!| ★

Directions: The following three questions refer to the graph below, which shows the change in the amount of DNA of a cell undergoing a certain type of cell division.



DNA replication starts at

A. P.
B. Q.
C. R.
D. S.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000065|!| ★

With reference to the graph, which of the following statements about this type of cell division are correct?

(1) These are two divisions.
(2) DNA replication occurs twice.
(3) The amount of DNA is halved at the end of the cell division.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000066|!| ★

Separation of DNA occurs at

A. R.

B. S.

C. R and S.

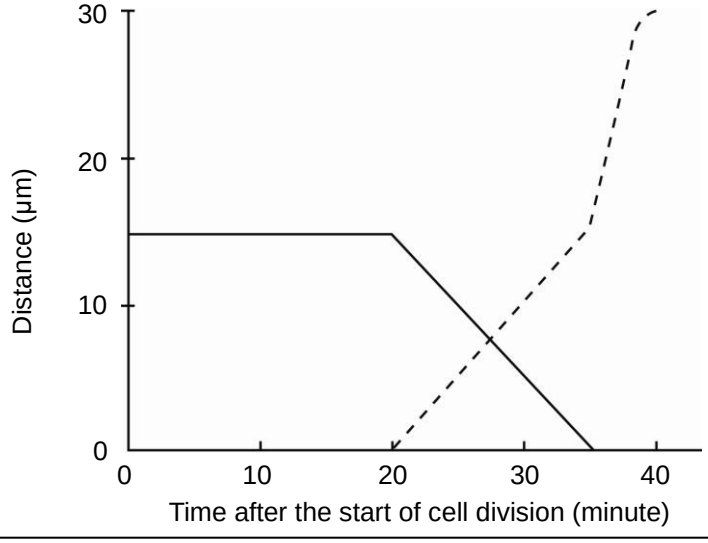
D. S and T.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0120000067|!| ★★

Directions: The following two questions refer to the graph below which is concerned with mitosis.



- Solid line represents the distance between daughter chromosomes and the poles towards which the chromosomes are moving.
- Broken line represents the distance between daughter chromosomes.

How long after the start of cell division do the chromosomes split into two daughter chromosomes and begin their movement towards the poles of the cell?

A. 0 minutes

B. 20 minutes

C. 27.5 minutes

D. 35 minutes

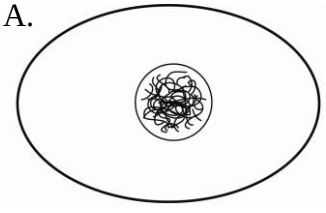
##B##

Skill assessed	Mark
Analysing data	1

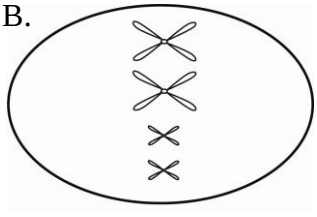
|!|BCEMA0120000068|!| ★★

Which of the following diagrams shows the arrangement of the chromosomes at 15 minutes?

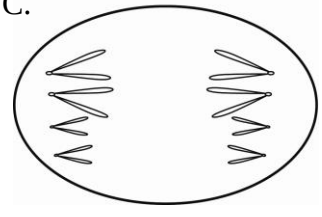
A.



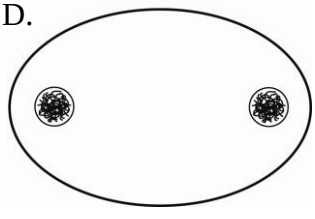
B.



C.



D.



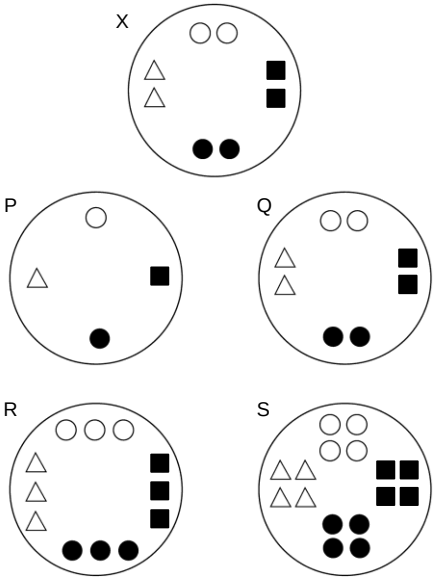
□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000069|!|

Directions: The following three questions refer to the diagrams below which show the chromosome patterns of the cells of the same species. X represents the normal homologous chromosome pairs in a body cell.



If a gamete was derived from X, which of the following represents the chromosome pattern of the gamete?

A. P
B. Q
C. R
D. S

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000070|!|

If X underwent mitotic cell division to produce some daughter cells, which of the following represents the chromosome pattern of the daughter cell?

A. P
B. Q
C. R
D. S

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0120000071|!|

Which of the following shows the chromosome pattern of the zygote that was formed from the fusion of two gametes if the gametes were derived from X?

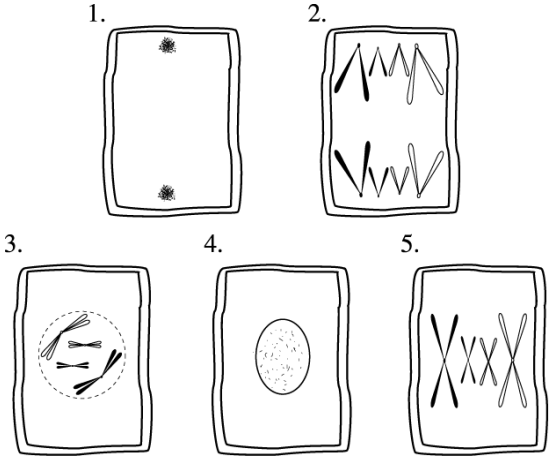
A. P
B. Q
C. R
D. S

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000072|!| ★

Directions: The following two questions refer to the diagrams below which show the stages of the division of a cell.



The correct sequence of the stages of cell division is

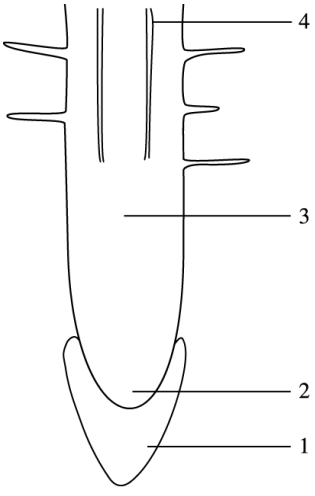
A. 1 → 2 → 5 → 3 → 4.
B. 3 → 4 → 5 → 2 → 1.
C. 4 → 5 → 2 → 1 → 3.
D. 4 → 3 → 5 → 2 → 1.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000073|!| ★ **Cross topic**

In which region of an onion root tip would such division be found?



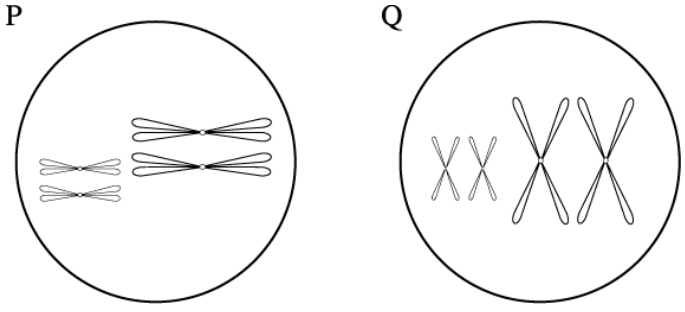
A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000074|!| ★

Directions: The following three questions refer to the diagrams below which show the stages of two types of nuclear division (P and Q) in animal cells.



Which of the following comparisons between nuclear divisions P and Q is ***incorrect***?

	Division P	***Division Q***
A.	Crossing over may occur.	No crossing over occurs.
B.	Homologous chromosomes pair up.	Homologous chromosomes do not pair up.
C.	It produces haploid daughter cells.	It produces diploid daughter cells.
D.	Sister chromatids separate to the opposite poles of the cell.	Homologous chromosomes separate to the opposite poles of the cell.

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000075|!| ★

Nuclear division P takes place in the

A. muscle.
B. ovary.
C. oviduct.
D. sperm duct.

□

##B##

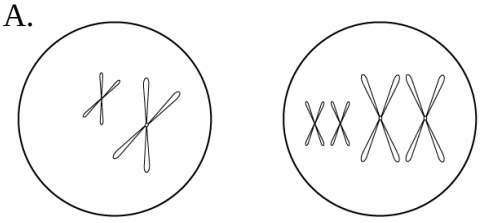
Skill assessed	Mark
Applying concepts	1

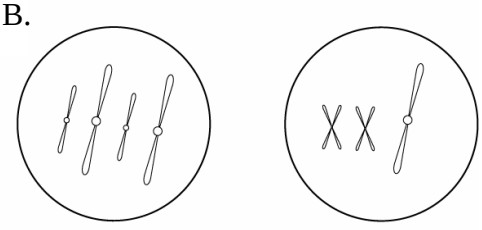
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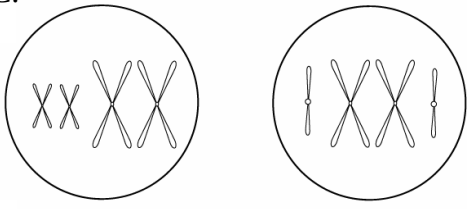
|!|BCEMB0120000076|!| ★

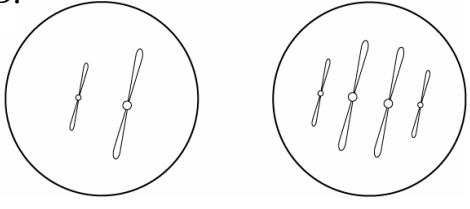
Which of the following pairs of daughter cells are formed at the end of each nuclear division?

P **Q**

A. 

B. 

C. 

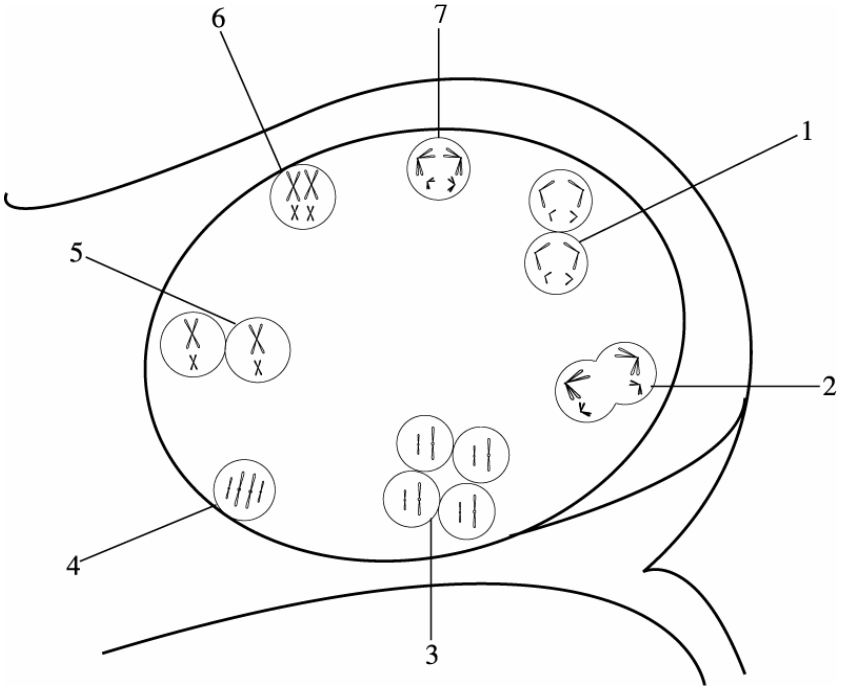
D. 

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000077|!| ★

Directions: The following three questions refer to the diagram below which shows the stages of cell division in certain part of a plant.



The correct sequence of the stages is

A. 4 → 6 → 7 → 1 → 5 → 2 → 3.

B. 4 → 6 → 7 → 2 → 5 → 1 → 3.

C. 6 → 7 → 4 → 1 → 5 → 2 → 3.

D. 6 → 7 → 4 → 5 → 1 → 2 → 3.

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000078|!| ★ **Cross topic**

The type of cell division shown in the diagram is

A. meiotic cell division in an anther.

B. meiotic cell division in the shoot tip.

C. mitotic cell division in the root tip.

D. mitotic cell division in the ovary.

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0120000079|!| ★

Which of the following indicate(s) the importance of this type of cell division to a flowering plant?

(1) Flowering plants can produce more gametes and increase the chance of pollination.
(2) The diploid number of chromosomes can be restored at fertilization.
(3) Flowering plants can produce genetically identical offspring.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

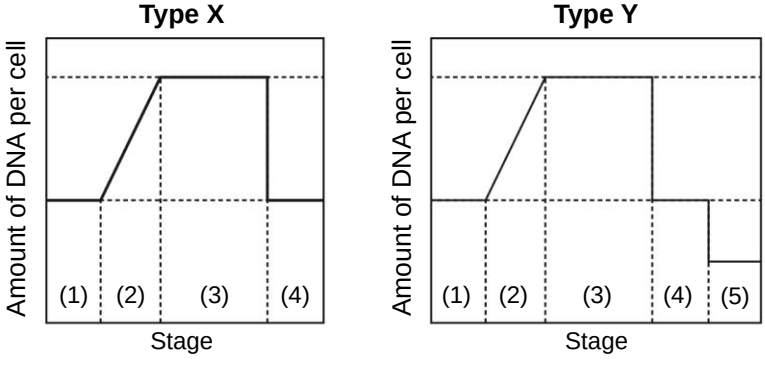
□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0120000080|!| ★★

Directions: The following three questions refer to the graphs below which show the changes in the amount of DNA during two types of cell division over a period of time.



Which of the following is ***not*** a correct comparison between these two types of cell division?

	Cell division type X	***Cell division type Y***
A.	chromosome number in the daughter cell is diploid	chromosome number in the daughter cell is haploid
B.	occurs in body cells	occurs in gamete-producing cells
C.	pairing of homologous chromosomes is not involved	pairing of homologous chromosomes is involved
D.	chromosomes form a double row along the equatorial plane of the cell	chromosomes form a single row along the equatorial plane of the cell

☐

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0120000081|!| ★★

Which of the following occurs in stage (2)?

A. DNA replication
B. separation of homologous chromosomes
C. separation of sister chromatids
D. uncoiling of chromosomes into chromatin

☐

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0120000082|!| ★★

Which of the following does ***not*** occur from stage (3) to (4) in type Y cell division?

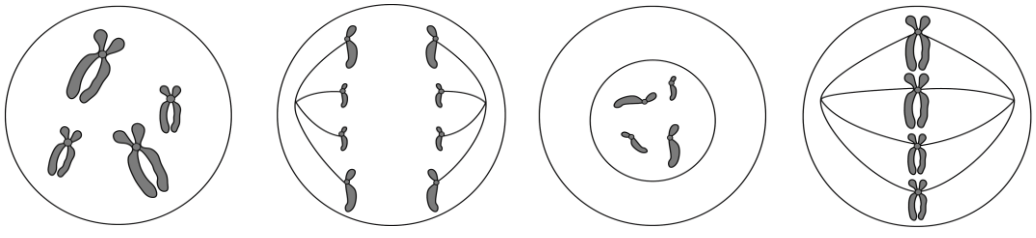
A. crossing over
B. synthesis of new organelles
C. separation of homologous chromosomes
D. degeneration of nuclear membrane

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000083|!| ★

Directions: The following two questions refer to the following diagram which shows four different stages of cell division in humans.



(1) **(2)** **(3)** **(4)**

Which of the following is the correct order of the stages?

A. (1) → (3) → (4) → (2)
B. (1) → (4) → (2) → (3)
C. (3) → (1) → (4) → (2)
D. (3) → (4) → (2) → (1)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0120000084|!| ★

This type of cell division takes place in

(1) the liver.
(2) the bone marrow.
(3) the epidermis of skin.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##D##

Skill assessed	Mark
Applying concepts	1